

Markov Chain

1. RWs.
2. Birth & Death chain
 - Ehrenfest model of Diffusion
 - Bernoulli Laplace model of
3. Queueing model Diffusion
4. Wright Fisher model.
5. Branching pr.

Bernoulli Laplace model.



Cont. A



Cont. B

Two types of particles.

Total no. of particles of each type is N .

No. of particles in each container is N .

Dynamics: - Pick one particle from each container and swap them.

X_n be the part. of type 1 in container A

$\{X_n: n \geq 0\}$ is a M.C.

$S = \{0, 1, \dots, N\}$

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & \dots & i-1 & i & i+1 & \dots & N \\ \hline 0 & 0 & 1 & & & & & & & \\ 1 & & & & & & & & & \\ 2 & & & & & & & & & \\ \vdots & & & & & & & & & \\ i & & & & & & & & & \\ \vdots & & & & & & & & & \\ N & & & & & & & & & \end{array}$$

$$\left(\frac{i}{N}\right)^2 \frac{i}{N} \left(\frac{N-i}{N}\right) \left(\frac{N-i}{N}\right)^2$$

$$1 \ 0 \ 1$$

$$p_{ij} = \begin{cases} 0 & \text{if } |i-j| > 1 \\ \frac{i}{N} \frac{(N-i)}{N} & j = i \\ \left(\frac{N-i}{N}\right)^2 & j = i+1 \\ \left(\frac{i}{N}\right)^2 & j = i-1 \end{cases}$$

Queueing model: $\rightarrow$ Think of taxi stand where at each time pt. a taxi comes and takes the first customer, if any,

and leaves. Between two time pts, n & $n+1$,

ξ_{n+1} many customers arrive.

Assume that $\{\xi_n : n \geq 1\}$ are iid.

Let $P(\xi_n = k) = \alpha_k, \quad k = 0, 1, \dots,$

X_n be the no. of customers waiting at time n ,

$$\sum_{k=0}^{\infty} \alpha_k = 1$$

If $X_n = 0$,

$$X_{n+1} = \max(0, \xi_{n+1} - 1)$$



If $X_n > 1$,

$$X_{n+1} = \xi_{n+1}$$

$$X_{n+1} = f(X_n, \xi_{n+1})$$

otherwise

$\Rightarrow \{X_n\}$ M.C.

$$X_{n+1} = (X_n - 1) + \xi_{n+1}$$

state sp. $\{0, 1, -\}$

$$p_{ij} = \begin{cases} \alpha_0 & i=0 \end{cases}$$

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & \dots & i-1 & i & i+1 & i+2 & \dots \\ \hline 0 & \alpha_0 + \alpha_1 & \alpha_2 & \alpha_3 & \dots & \alpha_i & \alpha_{i+1} & \alpha_{i+2} & & \\ 1 & \alpha_0 & \alpha_1 & \alpha_2 & & \alpha_{i-1} & \alpha_i & \alpha_{i+1} & & \\ 2 & 0 & \alpha_0 & \alpha_1 & \alpha_2 & & \alpha_{i-2} & \alpha_{i-1} & \alpha_i & \\ & & & & \vdots & & & & & \\ k & 0 & 0 & \dots & 0 & \alpha_0 & \alpha_1 & & & \\ & \underbrace{\hspace{10em}}_{k-1} & & & & & & & & \end{array}$$

Wright Fisher model.

Let X_n be the no.
of positions occupied by

a . The no of positions
with a follows

$$X_{n+1} \sim \text{Bin}\left(N, \frac{X_n}{N}\right).$$

a, A

N positions
where either

a or A is
placed.

$$\text{Bin}\left(N, \frac{X_n}{N}\right).$$

$$\begin{aligned}
 p_{ij} &= P(X_{n+1}=j \mid X_n=i) \\
 &= P(\text{Bin}(N, \frac{i}{N})=j) \\
 &= \binom{N}{j} \left(\frac{i}{N}\right)^j \left(1 - \frac{i}{N}\right)^{N-j}
 \end{aligned}$$

Observation :-

$$\begin{aligned}
 E(X_{n+1} \mid X_n=i) &= N \times \frac{i}{N} = i \\
 \Rightarrow E(X_{n+1} \mid X_n) &= X_n.
 \end{aligned}$$

Branching for.

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$$

$$X_{n+1}^2 f(X_n, E_{n+1})$$

$$\{\xi_i^{(n)} : i \geq 1, n \geq 1\}$$

$$f(k, E) = \begin{cases} \sum_{i=1}^k E_i \\ 0 \end{cases}$$

$$\text{if } k \geq 1. \quad E_{n+1} = \{\xi_i^{(n)} : i \geq 1\}$$

$$\Rightarrow Z_{n+1} = f(Z_n, E_{n+1})$$

$$\text{o.w. } E = (0, 1, 4, 6, 2, 0, 3, \dots)$$

$$p_{ij} = P(Z_{n+1}=j \mid Z_n=i)$$

$$= \text{Coeff. of } s^j \text{ in } (G(s))^i$$

Generic finite example.

$$S = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

$G(s)$ is the pgf
of $\mathbb{Z}$

$$P = \begin{array}{c|cccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline 0 & \frac{1}{4} & & & \frac{3}{4} & & & & \\ 1 & \frac{1}{2} & 0 & \frac{1}{3} & & & & \frac{1}{6} & \\ 2 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} \\ 3 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 5 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 6 & & \frac{1}{3} & & & & & & \\ 7 & & & \frac{1}{3} & & & \frac{1}{3} & & \frac{1}{3} \end{array}$$

Polya's Urn model

At time 0, an urn contains r many red balls and b many blue balls. Draw one at random, note the colour, replace it and add $c > 0$ many balls of the same color. Let X_n denote the colour of the ball at n th draw.

$\{X_n : n \geq 0\}$ is NOT a M.C. | $X_n \in \{R, B\}$
 $S = \{R, B\}$.

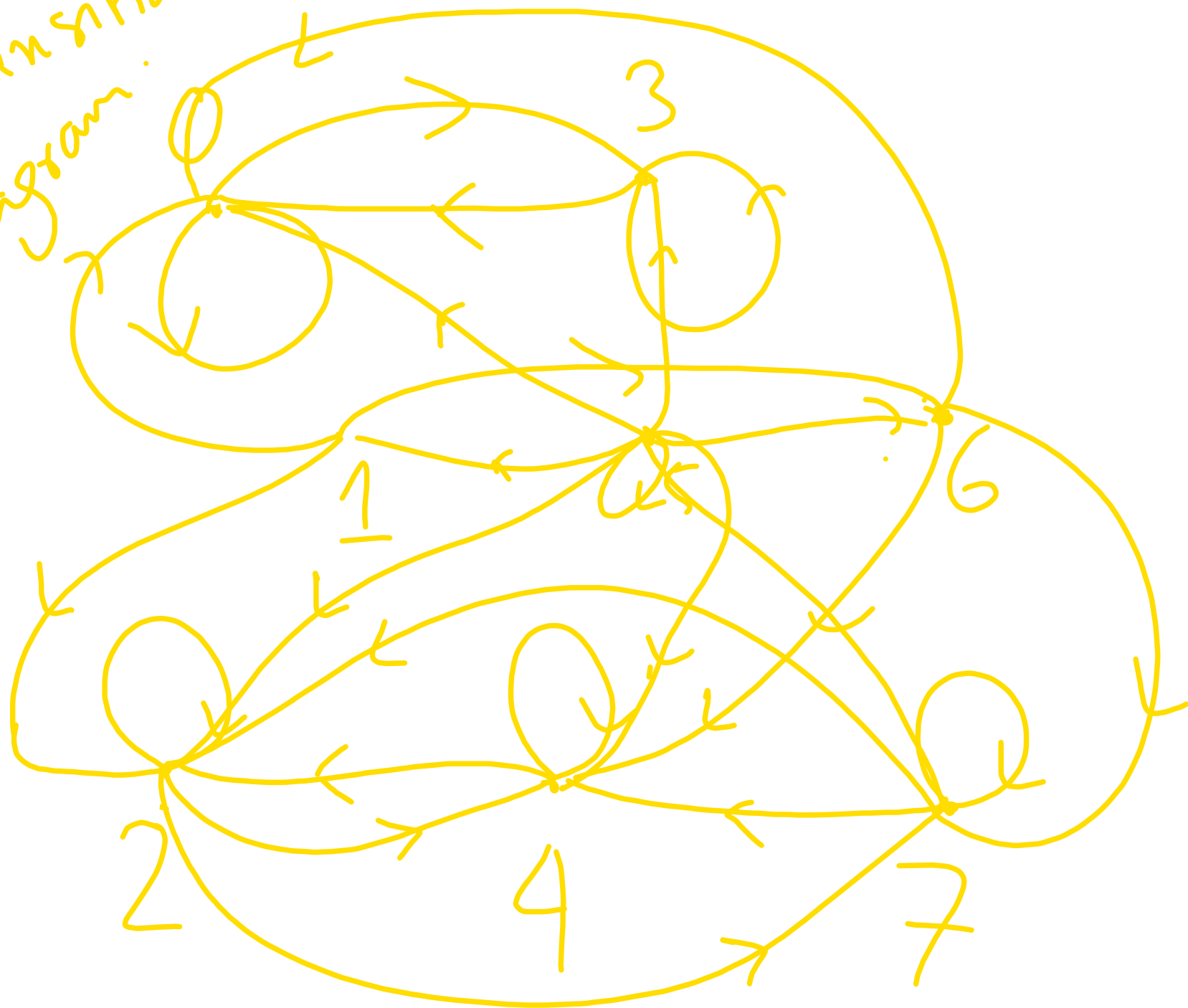
$$P(X_2 = R \mid X_1 = R), P(X_2 = R \mid X_1 = R, X_0 = B)$$

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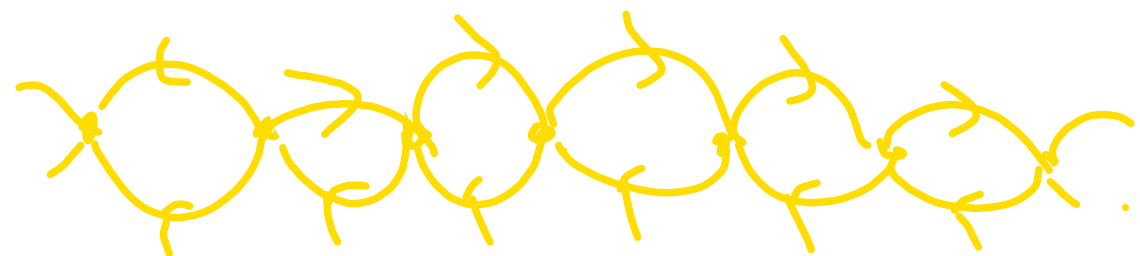
Transition diagrams.

$P = ((P_{ij})) \Rightarrow$ Directed graph.
with vertices as the states
space S , and edge from
 i to j if $P_{ij} > 0$.

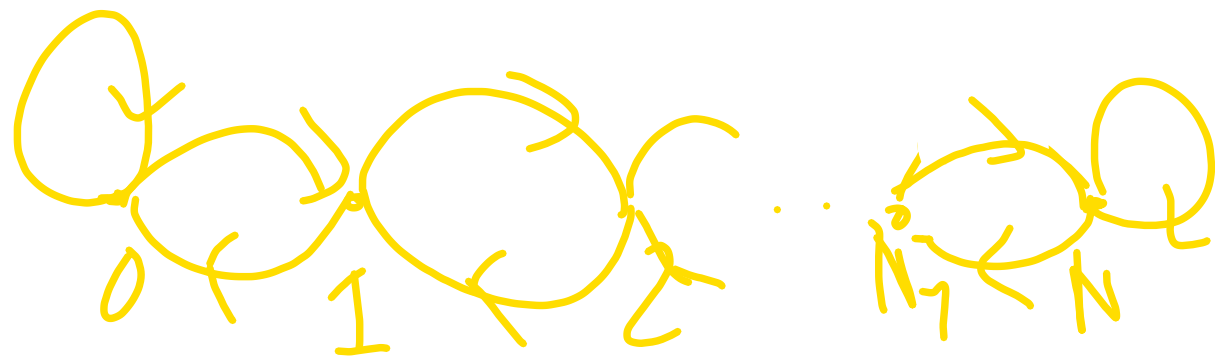
Transition
Diagram:



RW



Gambler's Ruin



Higher transitions.

$A_0, A_1, \dots, A_n.$

$$P(A_0 \cap A_1 \cap \dots \cap A_n)$$

$$= P(A_n | A_{n-1} \cap \dots \cap A_0) \times P(A_{n-1} \cap \dots \cap A_0)$$

$$= P(A_n | A_{n-1} \cap \dots \cap A_0) P(A_{n-1} | A_{n-2} \cap \dots \cap A_0) \\ \dots P(A_2 | A_1 \cap A_0) P(A_1 | A_0) P(A_0).$$

$$P(\underbrace{X_n = i_n}_{A_n}, \underbrace{X_{n-1} = i_{n-1}}_{A_{n-1}}, \dots, \underbrace{X_0 = i_0}_{A_0})$$

$$= P(X_n = i_n \mid X_{n-1} = i_{n-1}, \dots, X_0 = i_0)$$

$$P(X_{n-1} = i_{n-1} \mid X_{n-2} = i_{n-2}, \dots, X_0 = i_0)$$

⋮

$$P(X_2 = i_2 \mid X_1 = i_1, X_0 = i_0)$$

$$P(X_1 = i_1 \mid X_0 = i_0) P(X_0 = i_0)$$

$$= p_{0, n-1, i_n} p_{i_{n-2}, i_{n-1}} \dots p_{i_1, i_2} p_{i_0, i_1} P(X_0 = i_0)$$

$$P(X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_1 = i_1 | X_0 = i_0).$$

$$= \prod_{j=0}^{n-1} p_{i_j, i_{j+1}}$$